

SURAJ MISHRA

Senior DevOps Engineer · Kubernetes & Cloud Infrastructure · AWS / Azure · GPU Infrastructure & MLOps

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SUMMARY

Senior DevOps Engineer with **4.5+ years** of experience designing, automating and operating production infrastructure across **AWS, Azure and on-prem environments**. Strong hands-on expertise in **Kubernetes platform engineering, Docker, Terraform, CI/CD, GitOps, cloud infrastructure, observability, DevSecOps, high availability and disaster recovery**. Specialized experience building **GPU infrastructure for ML inference** using NVIDIA drivers/CUDA, NVIDIA GPU Operator, MIG partitioning, GPU time-slicing and KEDA-based autoscaling. Delivered measurable improvements including **55% lower GPU cost per workload, 45% lower idle GPU spend, 99.9% availability, 3-day to 4-hour environment provisioning and 3-hour to 15-minute release preparation**.

TECHNICAL SKILLS

Cloud	AWS (EKS, EC2, S3, Lambda, VPC, IAM, CloudFormation) · Azure (AKS, VNet) · GCP · On-Prem
Kubernetes	Kubernetes · Docker · Helm · KEDA · HPA · Cluster Autoscaler · RBAC · Workload Management · Production Operations
GPU / MLOps	NVIDIA Drivers & CUDA · NVIDIA GPU Operator · MIG Partitioning · GPU Time-Slicing · GPU Workload Sharing · Model Deployment · ML Inference Autoscaling
IaC / Automation	Terraform · Ansible · Reusable Infrastructure Modules · Configuration Management · Infrastructure Automation
CI/CD / GitOps	GitLab CI/CD · Jenkins · GitHub Actions · ArgoCD · GitOps · Automated Build / Test / Release Pipelines
DevSecOps	SonarQube · Snyk · Trivy · SAST/DAST Gating · Container Image Scanning · Security Gates · Automated Security Validation
Observability / SRE	Prometheus · Grafana · Alertmanager · Metrics · Dashboards · Alerting · SLI/SLO · Incident Response · MTTD · Production Monitoring
Reliability	High Availability · DC-DR · Automated Failover · Database Replication · Backup / Recovery · Production Reliability · Cost Optimization
Cloud Networking	AWS VPC · Azure VNet · Nginx · Linux Networking · Production Traffic Management
Databases	ScyllaDB · MongoDB · Redis · Replication · Sharding · Automated Backups · Multi-Region Operations
Systems / Programming	Linux · Bash · Python · Git · Nginx

EXPERIENCE

DevOps Engineer II — PrivaSapien Technologies

Jun 2025 – Present

Senior DevOps / Multi-Cloud, Kubernetes, GPU & Platform Infrastructure

Gurugram, India

- Architected and delivered production infrastructure for an enterprise consent-management platform supporting a **Fortune 500 automotive client**, integrating workloads across **AWS and Azure** with **zero downtime** during production integration.
- Engineered production **GPU infrastructure** using NVIDIA drivers/CUDA, NVIDIA GPU Operator, **MIG partitioning** and GPU time-slicing, increasing workload density to **6 workloads per node** and reducing per-workload GPU cost by **55%**.
- Designed **KEDA-based ML inference autoscaling** driven by live GPU utilisation metrics, reducing idle GPU spend by **45%** while maintaining **180ms p95** inference latency.
- Standardised Kubernetes application delivery through a reusable, parameterised **Helm chart library**, replacing **12 environment-specific chart sets** across development, staging, UAT and production and reducing release preparation from **3 hours to 15 minutes**.
- Built reusable **Terraform modules** for AWS, Azure and on-prem infrastructure, reducing new-environment provisioning from **3 days to 4 hours** and reducing infrastructure configuration drift.
- Improved production observability and reliability using **Prometheus, Grafana and SLO-based monitoring** for

GPU utilisation, model latency and Kubernetes cluster health, reducing mean time to detection from **25 minutes to under 5 minutes**.

- Implemented **DevSecOps controls** across the CI/CD lifecycle by enforcing SonarQube, Snyk, Trivy and automated testing as **blocking merge gates**, eliminating manual security approval from the release path.

DevOps Engineer — Sinch India

Mar 2022 – May 2025

Kubernetes, AWS, CI/CD & Production Infrastructure

Noida, India

- Operated and improved high-availability **Kubernetes and Docker** infrastructure supporting **25+ microservices**, reducing deployment time by **40%** and improving system efficiency by **30%**.
- Designed and supported a production **DC-DR architecture** with automated failover and replicated database clusters, sustaining **99.9% uptime** and reducing manual recovery effort by **70%**.
- Automated AWS infrastructure provisioning using **Terraform and CloudFormation** across **5 environments**, combining infrastructure automation, resource rightsizing and S3 lifecycle policies to reduce monthly cloud spend by **25%**.
- Built and maintained **GitLab CI/CD pipelines** for automated build, test and release workflows across **25+ services**, reducing deployment time by a further **50%**.
- Administered production **ScyllaDB, MongoDB and Redis** clusters across multi-region deployments, including replication, sharding, automated backups and operational recovery.
- Mentored **4 engineers** on Kubernetes and DevOps practices, contributing to branching strategy, code review standards and engineering consistency across teams.

KEY PROJECTS

- **GPU Sharing & ML Inference Platform** — Engineered GPU workload sharing using **MIG partitioning and time-slicing** and implemented **KEDA-based autoscaling** using live GPU metrics. Increased average GPU utilisation from **30% to 78%**, improving workload density and infrastructure efficiency.
- **Multi-Cloud Infrastructure & Reliability Platform** — Standardised infrastructure across **AWS, Azure and on-prem** environments using reusable **Terraform modules**, Kubernetes and Helm. Combined infrastructure automation, production monitoring, SLO-based observability, disaster recovery and cost optimisation to reduce provisioning time and operational overhead.

EDUCATION

B.Tech, Computer Science & Engineering — Lovely Professional University, Jalandhar

2018 – 2022

CGPA: 7.69 / 10